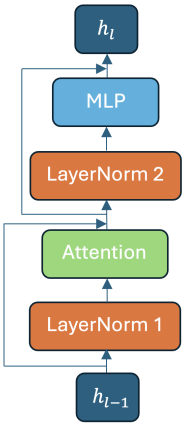


Figure 6: A high-level diagram representing the computation within one decoder block of an LLM.

A.2 Hyperparameters and Computation Resources

This appendix includes detailed tables and figures that illustrate the effects of layer normalization on the hidden and post-layer normalization vectors across various models.

All experiments were conducted using NVIDIA A6000 GPUs with 48GB of GPU memory. This setup provided the necessary computational power to handle large-scale models and extensive data processing required for our study.

Plots in Figure 7 compare the norms of the residual streams before and after LN1 across the rest of the models not included in Figure 4, demonstrating the normalization's effect. Plots in Figure 8, on the other hand, display the effect of LN2 on the residual stream norms across all models, highlighting the further stabilization achieved. Figure 9 shows the rotation angles between hidden vectors and post-LN1 vectors for the remaining models, illustrating the angular changes due to the first layer normalization. Figure 10 presents the rotation angles between hidden vectors and post-LN2 vectors for all models, and Figure 11 presents the angle frequency plots for the rest of the models.

A.3 Additional Ablations

Model Name	Hidden & Uniform (Mean ± Std)	Post-LN1 & Uniform (Mean ± Std)	Post-LN2 & Uniform (Mean ± Std)
GPT-2 XL	101.48 ± 3.20	89.95 ± 1.30	89.63 ± 1.30
GPT-Neo 1.3B	89.63 ± 1.84	88.85 ± 1.82	89.47 ± 1.84
Pythia-1.4B	89.88 ± 1.84	88.74 ± 1.81	89.39 ± 1.83
GPT-J 6B	88.74 ± 1.68	89.59 ± 1.69	-
Pythia-6.9B	89.27 ± 1.58	89.02 ± 1.57	88.97 ± 1.57
Llama-2-7B-hf	89.47 ± 0.15	89.48 ± 0.14	89.72 ± 0.15
Llama-3-8B	90.01 ± 0.15	90.15 ± 0.15	89.97 ± 0.15

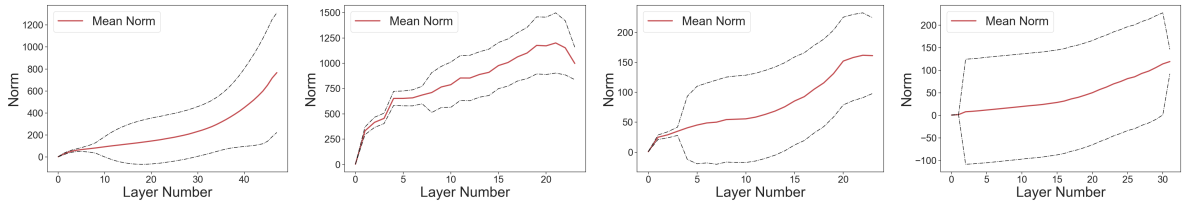
Table 2: Average angles in degrees between a uniform vector and vectors from Hidden, Post-LN1, and Post-LN2 layers across all layers for each model

Model Name	Hidden (Mean ± Std)	Max Hidden	Post-LN1 (Mean ± Std)	Max Post-LN1	Post-LN2 (Mean ± Std)	Max Post-LN2
GPT-2 XL	236.85 ± 37.02	7186.44	21.42 ± 0.47	37.42	23.79 ± 0.49	38.51
GPT-Neo 1.3B	811.84 ± 40.29	5961.95	6.69 ± 0.26	32.99	13.79 ± 0.49	50.49
Pythia-1.4B	77.94 ± 13.17	1373.85	46.02 ± 1.11	81.54	35.68 ± 1.13	56.53
GPT-J 6B	110.88 ± 30.02	4757.86	49.44 ± 1.31	70.93	-	-
Pythia-6.9B	216.04 ± 24.61	3119.49	79.65 ± 1.56	133.55	42.18 ± 1.03	65.39
Llama-2-7B-hf	44.30 ± 19.55	3287.86	26.21 ± 0.56	38.06	17.27 ± 0.42	30.76
Llama-3-8B	12.12 ± 3.15	549.19	29.46 ± 0.59	46.21	21.30 ± 0.49	39.67

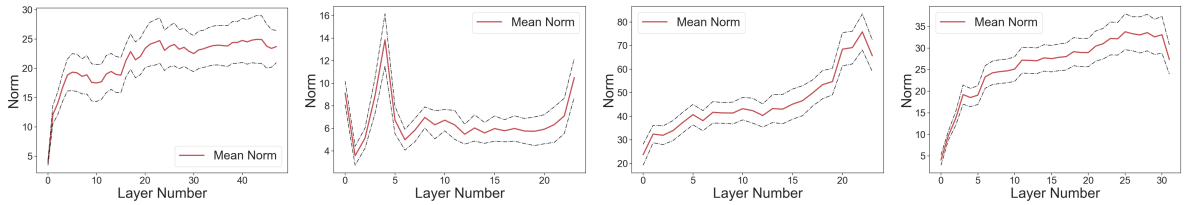
Table 3: This table presents the average norms (L2) and standard deviations for hidden input vectors and post-layer normalization output vectors, alongside the maximum observed values for various models. "Hidden" is an abbreviation for hidden input vectors.

Model Name	Hidden & Post-LN1 (Mean ± Std)	Hidden & Post-LN2 (Mean ± Std)
GPT-2 XL	30.46 ± 1.68	32.68 ± 1.60
GPT-Neo 1.3B	49.09 ± 1.22	52.82 ± 1.34
Pythia-1.4B	12.35 ± 0.46	21.70 ± 0.90
GPT-J 6B	19.55 ± 0.98	-
Pythia-6.9B	11.02 ± 0.26	28.66 ± 0.58
Llama-2-7B-hf	17.13 ± 0.77	23.48 ± 1.16
Llama-3-8B	19.64 ± 0.62	24.55 ± 1.08

Table 4: This table shows the average rotation angles in degrees between hidden and post-layer normalization vectors (LN1 and LN2) across all layers for various models, indicating the typical angular deviation introduced by normalization processes.

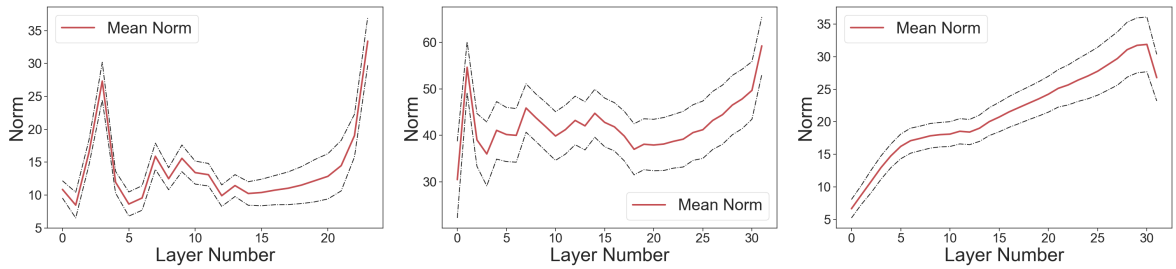


(a) Residual Stream Pre-LN1 GPT-2 XL (b) Residual Stream Pre-LN1 GPT-Neo (c) Residual Stream Pre-LN1 Pythia 1.4 (d) Residual Stream Pre-LN1 Llama-2

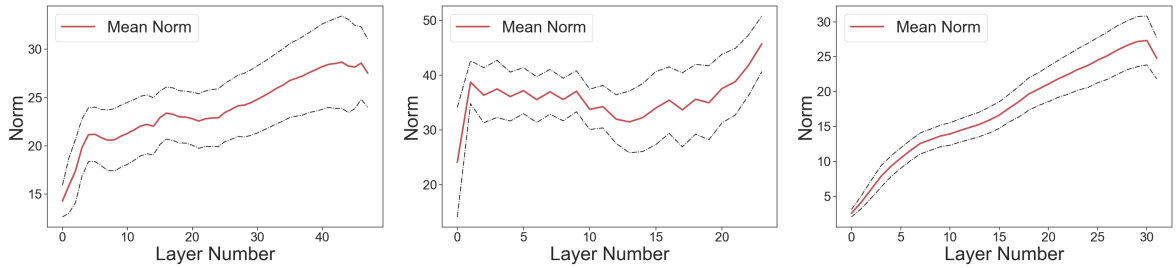


(e) Residual Stream Post-LN1 GPT-2 XL (f) Residual Stream Post-LN1 GPT-Neo (g) Residual Stream Post-LN1 Pythia 1.4 (h) Residual Stream Post-LN1 Llama-2

Figure 7: Pre and Post-LN1 plots of GPT-2 XL, GPT-Neo, Pythia 1.4 and Llama-2

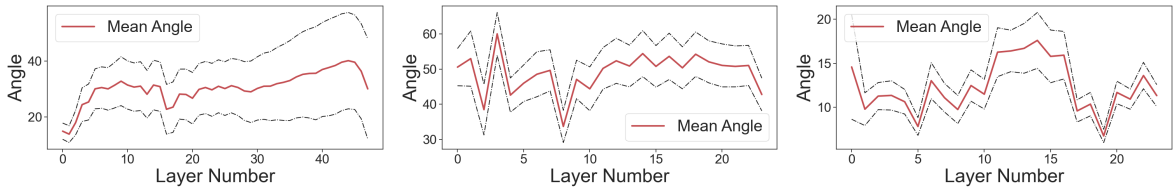


(a) Residual Stream Post-LN2 GPT-Neo (b) Residual Stream Post-LN2 Pythia 6.9 (c) Residual Stream Post-LN2 Llama-3

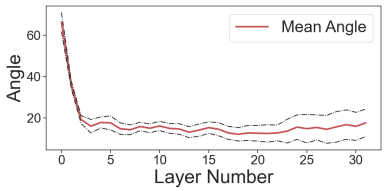


(d) Residual Stream Post-LN2 GPT-2 XL (e) Residual Stream Post-LN2 Pythia 1.4 (f) Residual Stream Post-LN2 Llama-2

Figure 8: Affect of LN2 on stabilizing the norm in all models



(a) GPT-2 XL (b) GPT-Neo (c) Pythia 1.4



(d) Llama-2

Figure 9: Rotation angle (in degrees) between the hidden vectors and Post-LN1 vectors across all layers for GPT-2 XL, GPT-Neo, Pythia 1.4 and Llama-2